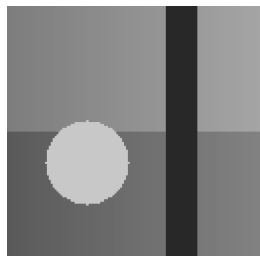
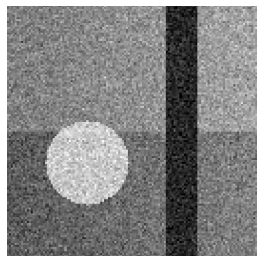


Image Denoising via a Gaussian Markov Random Field (GMRF) Prior

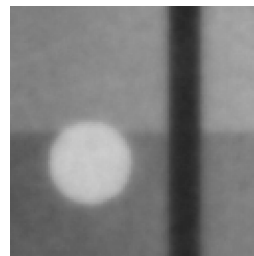
(a) Clean scene



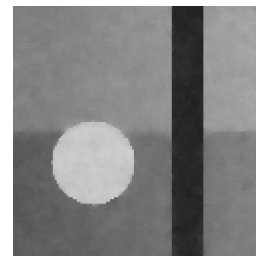
(b) Noisy, $\sigma=20$
PSNR=22.1 dB



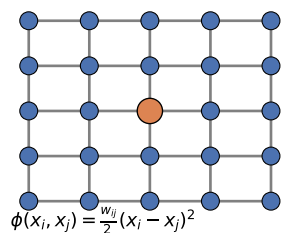
(c) Isotropic GMRF ($\lambda=6.0$)
PSNR=26.9 dB



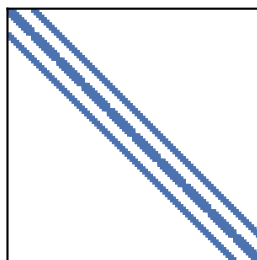
(d) Edge-preserving GMRF
PSNR=35.7 dB



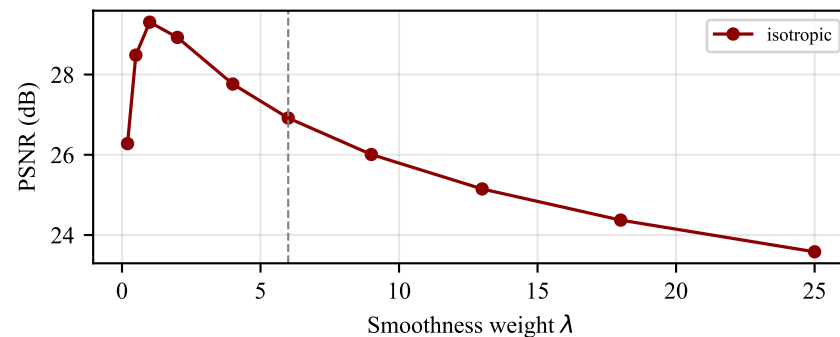
(e) 4-connected MRF graph



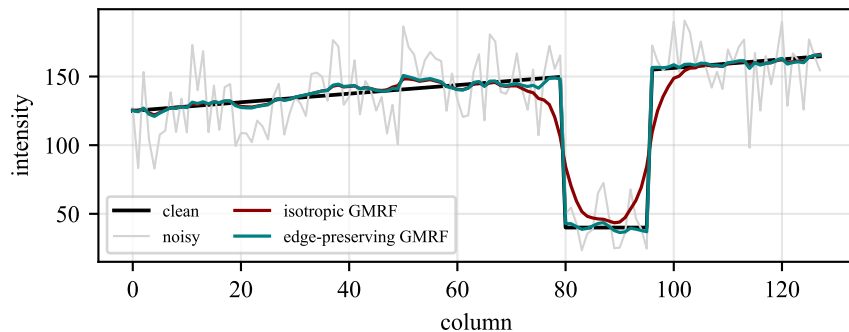
(f) Precision matrix
sparsity (10x10 patch)



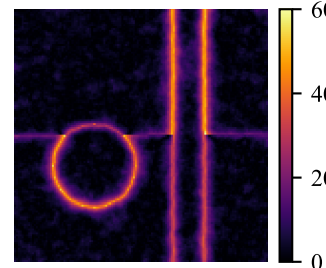
(g) Bias-variance trade-off over λ



(h) Row 40 profile: edge sharpness



(i) |error| isotropic



(i) |error| edge-preserving

